

Software Engineering (ESC501)

Course Details

Credit Points: 3

Syllabus Outline

Unit 1

Overview of System Analysis & Design, Business System Concept, System Development Life Cycle, Waterfall Model, Spiral Model, Feasibility Analysis, Technical Feasibility, Cost- Benefit Analysis, COCOMO model.

Unit 2

System Design - Context diagram and DFD, Problem Partitioning, Top-Down And Bottom-Up design; Decision tree, decision table and structured English; Functional vs. Object-Oriented approach.

Unit 3

Coding & Documentation - Structured Programming, OO Programming, Information Hiding, Reuse, System Documentation. Testing - Levels of Testing, Integration Testing, Test case Specification, Reliability Assessment, Validation & Verification Metrics, Monitoring & Control.

Unit 4

Software Project Management - Project Scheduling, Staffing, Software Configuration Management, Quality Assurance, Project Monitoring.

Unit 5

Static and dynamic models, why modeling, UML diagrams: Class diagram, interaction diagram: collaboration diagram, sequence diagram, state chart diagram, activity diagram, implementation diagram.

Software Engineering Lab (ESC591)

Course Details

Credit Points: 2

Laboratory Experiments

- Problem Analysis and Project Planning -Thorough study of the problem - Identify Project scope, Objectives and Infrastructure.
- Software Requirement Analysis - Describe the individual Phases/modules of the project and Identify deliverables. Identify functional and non-functional requirements.
- Data Modeling - Use work products - data dictionary.
- Software Designing - Develop use case diagrams and activity diagrams, build and test class diagrams, sequence diagrams and add interface to class diagrams.
- Prototype model - Develop the prototype of the product.
- The SRS and prototype model should be submitted for end semester examination.